

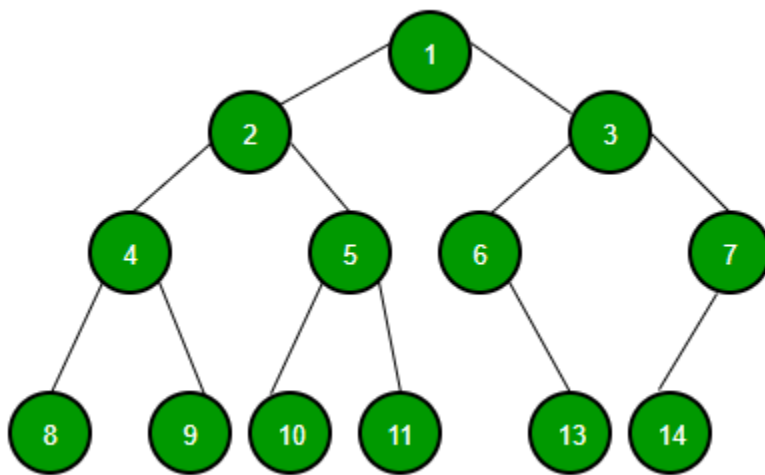
Name:

Section:

Part 1: The topic is Tree.

Question number 1: Define the following tree traversal, Inorder Traversal, Preorder Traversal, and Postorder Traversal.

Question number 2: Given the sample Binary tree below, show the nodes of the tree visited in each of the traversal techniques.



Inorder Traversal:
Preorder Traversal:
Postorder Traversal:

Question number 3: What is Binary Search Tree (BST)?

Question number 4: Given the following values, draw the BST.

30, 20, 25, 17, 48, 39, 53

Part 1: The topic is Graph.

Question number 1: Give the formal definition of Graph.

Q2: Given the following vertices and edges, draw the graph.

$$V = \{ a, b, c, d, e, f, g \}$$
$$E = \{ ac, ab, ad, bd, be, cd, de \}$$

Q3: Define Depth First Search and Breadth First Search.

Q4: Given the graph below, determine the output when applying Depth First Search and Breadth First Search.

